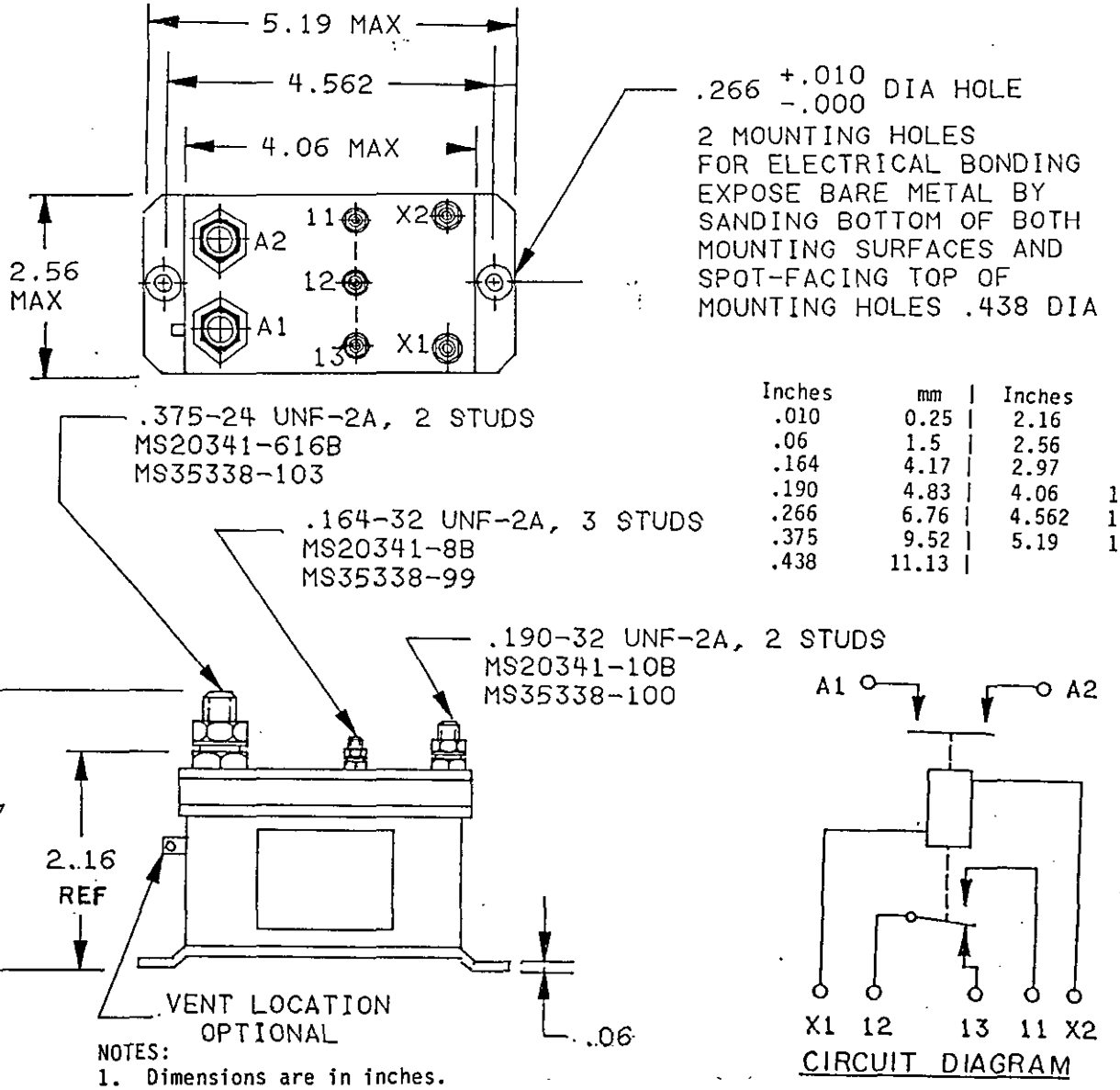
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE II (UNSEALED) 300 AMPERES, SPST (N.O.) WITH 5 AMPERE AUXILIARIES

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-6106.



Inches	mm	Inches	mm
.010	0.25	2.16	54.9
.06	1.5	2.56	65.0
.164	4.17	2.97	75.4
.190	4.83	4.06	103.1
.266	6.76	4.562	115.87
.375	9.52	5.19	131.8
.438	11.13		

CIRCUIT DIAGRAM

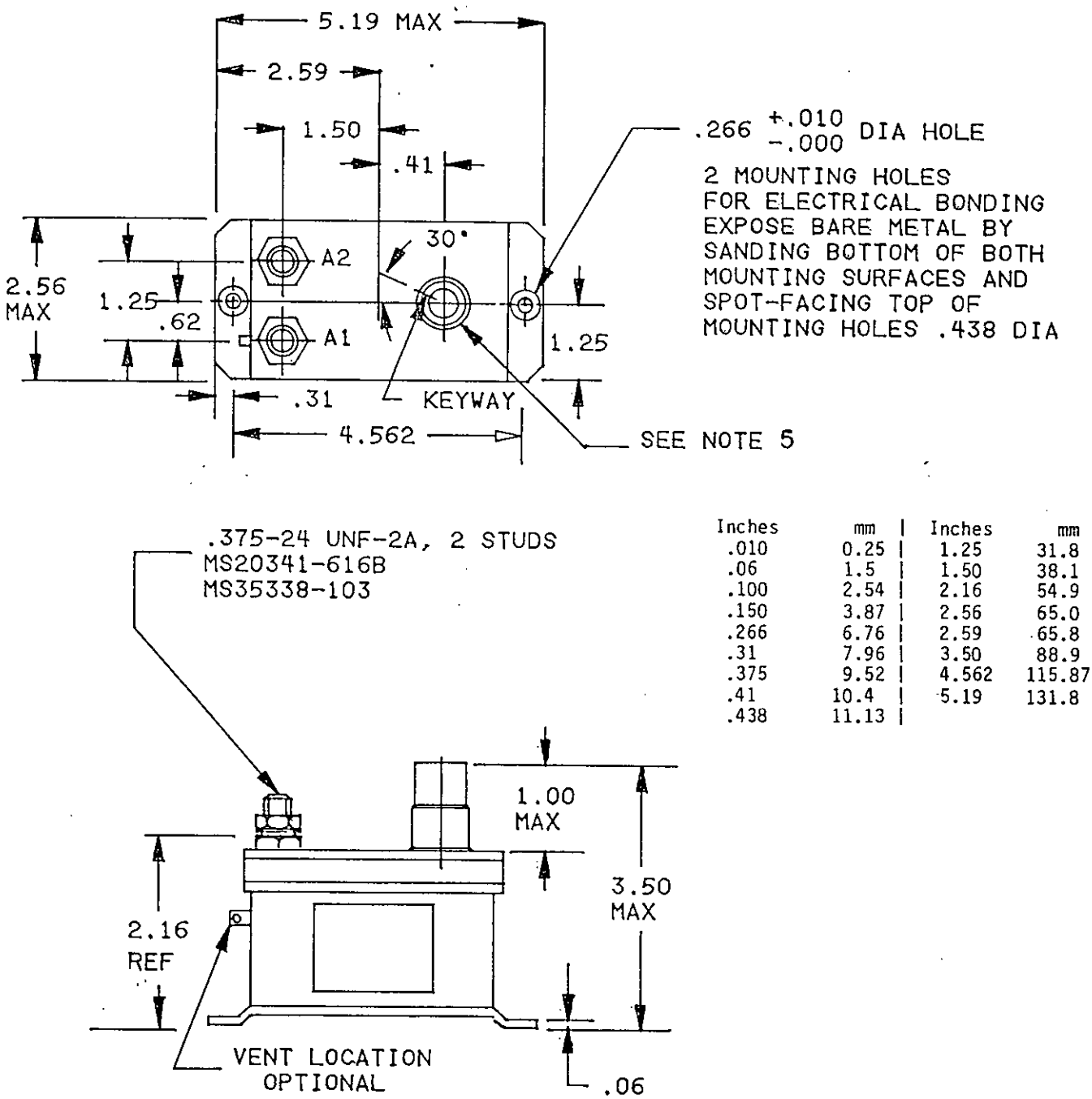
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.

Ⓑ FIGURE 1. Dimensions and configuration (-001 only).

Ⓑ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

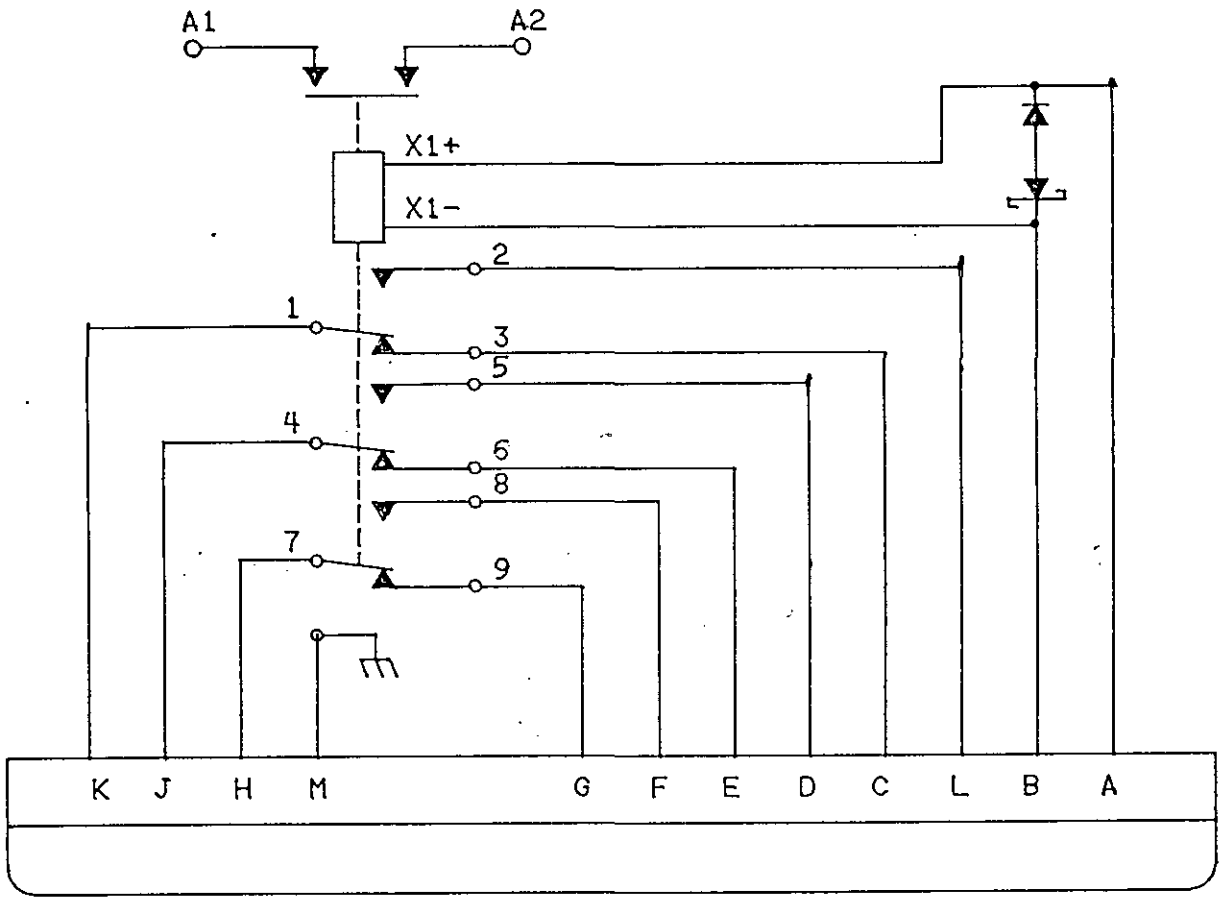
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Inches	mm	Inches	mm
.010	0.25	1.25	31.8
.06	1.5	1.50	38.1
.100	2.54	2.16	54.9
.150	3.87	2.56	65.0
.266	6.76	2.59	65.8
.31	7.96	3.50	88.9
.375	9.52	4.562	115.87
.41	10.4	5.19	131.8
.438	11.13		

Ⓑ FIGURE 2. Dimensions and configuration (-002 only).

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CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. Connector shall be MS3440H14C19P or equivalent.
6. Diodes shall have a peak inverse voltage at 600 V dc minimum.

Ⓑ FIGURE 2. Dimensions and configuration (-002 only). - Continued.

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REQUIREMENTS:

COIL DATA AND OPERATION DATA: See table I.

Coil data:

Duty: Continuous.

OPERATIONAL DATA:

Contact load: See table II.

(B) Contact bounce time: 15 milliseconds, maximum (primary contacts).
4.0 milliseconds, maximum (auxiliary contacts).

Contact drop: 150 millivolts, maximum, initially.
175 millivolts, maximum, after life test.

Overload current: 2,400 amperes (primary contacts).
20 amperes dc, 30 amperes ac (auxiliary contacts).

Rupture current: 3,000 amperes (primary contacts).
25 amperes dc, 40 amperes ac (auxiliary contacts).

PHYSICAL DATA:

Weight: 2 pounds maximum.

(B) Terminal strength: Applicable (except tested at room ambient temperature only).

Finish: Black.

Seal: Gasket sealed (vented).

Terminal barriers:

Base cover: Glass mat polyester per MIL-M-14.

Can: Aluminum.

ENVIRONMENTAL DATA:

Temperature range: -55°C to 71°C.

Altitude: 50,000 feet.

Dielectric withstanding voltage:

Sea level

	Initial	After life
Coil to case	1,050 V rms	1,000 V rms
Aux contacts	1,050 V rms	1,000 V rms
All other points	1,050 V rms	1,000 V rms

50,000 feet

Coil to case	500 V rms	500 V rms
Aux contacts	500 V rms	500 V rms
All other points	500 V rms	500 V rms

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Ⓑ TABLE I. Coil data and operational data.

Dash numbers M6106/34-	Rated			Max		Over temperature range					At 25°C			Time milliseconds max	
						Max pickup voltage			Hold voltage	Pickup voltage	Max pickup voltage	Hold voltage	Drop out voltage	Operate 2/	Release 3/
	Volts (V dc) 1/	Freq. Hz.	Res. Ω±10% @25°C	Volts (V dc)	Ampere	Normal (V dc)	High temp test (V dc)	Cont. current test (V dc)	(V dc)	(V dc)	(V dc)	(V dc)	(V dc)		
001	28	N/A	91	29	0.4	18	19	21	7.0	1.5	16	5.5	1.5	50	50
002	28	N/A	91	32	0.5	18	19	21	7.0	1.5	16	5.5	1.5	35	35

1/ CAUTION: Use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
2/ With rated coil voltage.
3/ From rated coil voltage.

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac 3 phase			
		Main		Aux		Main(Hz)		Aux(Hz)		Main(Hz)		Aux(Hz)	
		NO	NC	NO	NC	400	50/60	400	50/60	400	50/60	400	50/60
Resistive	50	300	---	5	5	---	---	5	---	---	---	---	---
Inductive	50	100	---	5	5	---	---	5	---	---	---	---	---
Inductive	---	---	---	---	---	---	---	---	---	---	---	---	---
Motor	50	250	---	---	---	---	---	---	---	---	---	---	---
Lamp	---	---	---	3	3	---	---	3	---	---	---	---	---
Transfer, load	---	---	---	---	---	---	---	---	---	---	---	---	---
Mechanical life	---	---	---	---	---	---	---	---	---	---	---	---	---
Reduced current	---	---	---	---	---	---	---	---	---	---	---	---	---
Inter. current		APPLICABLE (Auxiliary contacts only)											

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Insulation resistance: 100 megohms minimum at 500 volts.

Vibration (sinusoidal): .06 D.A., 10 to 57 Hz, 10 g's, 57 to 500 Hz.

Shock: 25 g's for 6 milliseconds (half-sine).

Acceleration: 10 g's.

Part number: M6106/34-(dash number from table I).

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F653)

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